
Labwork 4.2 - Recap

General Information and Submission Rules:

- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork4.c**. In addition to this, you need to create a **command line history** file with this command:
history > history_fullname_labwork4.txt
and submit this also.
- Incorrect naming or file format will result in a loss of 5 points from your total grade.
- Only use concepts covered in the labs so far.
- There is a bonus question in the end, it will add 10 points to your current grade. If your current grade is 100. These 10 points will be used in your next labwork as a bonus.

Question 1 - Days In Between Dates (25p)

- Write a C function (**calculate_dates**) that calculates the total day between two given dates. Pay attention to leap years.
- In addition it finds out the following day of the first date.

```
>> Enter first date: 01 03 2024
>> Enter second date: 04 04 2024
Day number: 34
Following day: 02 03 2024
```

Question 2 - Weather Guessing (25p)

- Create a function (**guess_weather**) that takes a temperature in degrees Celsius (an integer) from the user as input.
- Based on the following ranges, the function should print a weather guess:
 - Below 0 degrees: "Freezing cold!"
 - 0 to 10 degrees: "Very cold!"
 - 11 to 20 degrees: "Chilly!"
 - 21 to 30 degrees: "Warm!"
 - Above 30 degrees: "Hot!"

```
>> Enter the temperature: 5  
Very cold!
```

Question 3 - Print Pattern (25p)

- Write a function (**triangle**) that takes a digit from the user (from 1 to 9) and then prints a triangle pattern as given below.
- You can create two different functions (**up_triangle** and **down_triangle**) as well.

```
>> Enter a number: 5  
1  
12  
123  
1234  
12345  
1234  
123  
12  
1
```

Question 4 - Phone Number (25p)

- Create a new function (**phone_number**) that takes a phone number and a delimiter as an input from the user.
- Then by using that delimiter prints the phone number to the user as given.

```
>> Enter a phone number in given form (5XXXXXXXX): 5483920919  
>> Enter the delimiter: -  
548-392-09-19
```

Bonus: Greatest Common Divisor (GCD)

- The greatest common divisor (GCD) of integers a and b, at least one of which is nonzero, is the greatest positive integer d such that d is a divisor of both a and b.
- Write a function (**gcd**) that takes two inputs (two integers) from the user and prints the greatest common divisor of those numbers.

```
>> Enter two numbers: 36 60  
The gcd is 12.
```